

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular End Semester Examination – Winter 2022 Course: B. Tech. Branch :Chemical Engineering Semester :III Subject Code & Name: BTCHC304- Mechanical Operations Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the need of size reduction in Chemical process industries.	II/CO1	6
B)	Calculate the operating speed of the ball mill from the following data 1)Diameter of ball mill=800mm. 2) Diameter of balls = 60mm If Operating speed is 55% less than critical speed	VI/CO1	6
C)	Explain the working of Ribbon blender with well labeled diagram.	II/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain are the different types of Screen Analysis with its example in Tabular form.	II/CO2	6
B)	Explain the working of Vibrating Screen with well labeled diagram.	II/CO2	6
C)	Which are the factors which reduce capacity and effectiveness screen.	III/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Describe with neat sketch working of rotary drum filter.	II/CO3	6
B)	Explain the operating cycle of centrifuge.	II/CO3	6
C)	A plate and frame press, filtering a slurry gave a total of 8 m ³ of filtrate in 1800 sec and 11m ³ in 3066 sec, when filtration stopped. Estimate the washing time if 3 m ³ of wash water used. The resistance of the cloth can be neglected and a constant pressure is used throughout.	VI/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain Membrane processes based on size of component in the feed.	II/CO4	6
B)	Explain Ultrafiltration with its application, process driving force and membrane material used.	II/CO4	6
C)	Define Membrane separation processes with its industrial application over advantages over other unit operations.	I/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the process of batch sedimentation with diagram.	II/CO5	6
B)	With a neat sketch, describe the working of a DORR THICKENER.		6
C)	With suitable examples, explain the role of coagulants in filtration.	II/CO5	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.